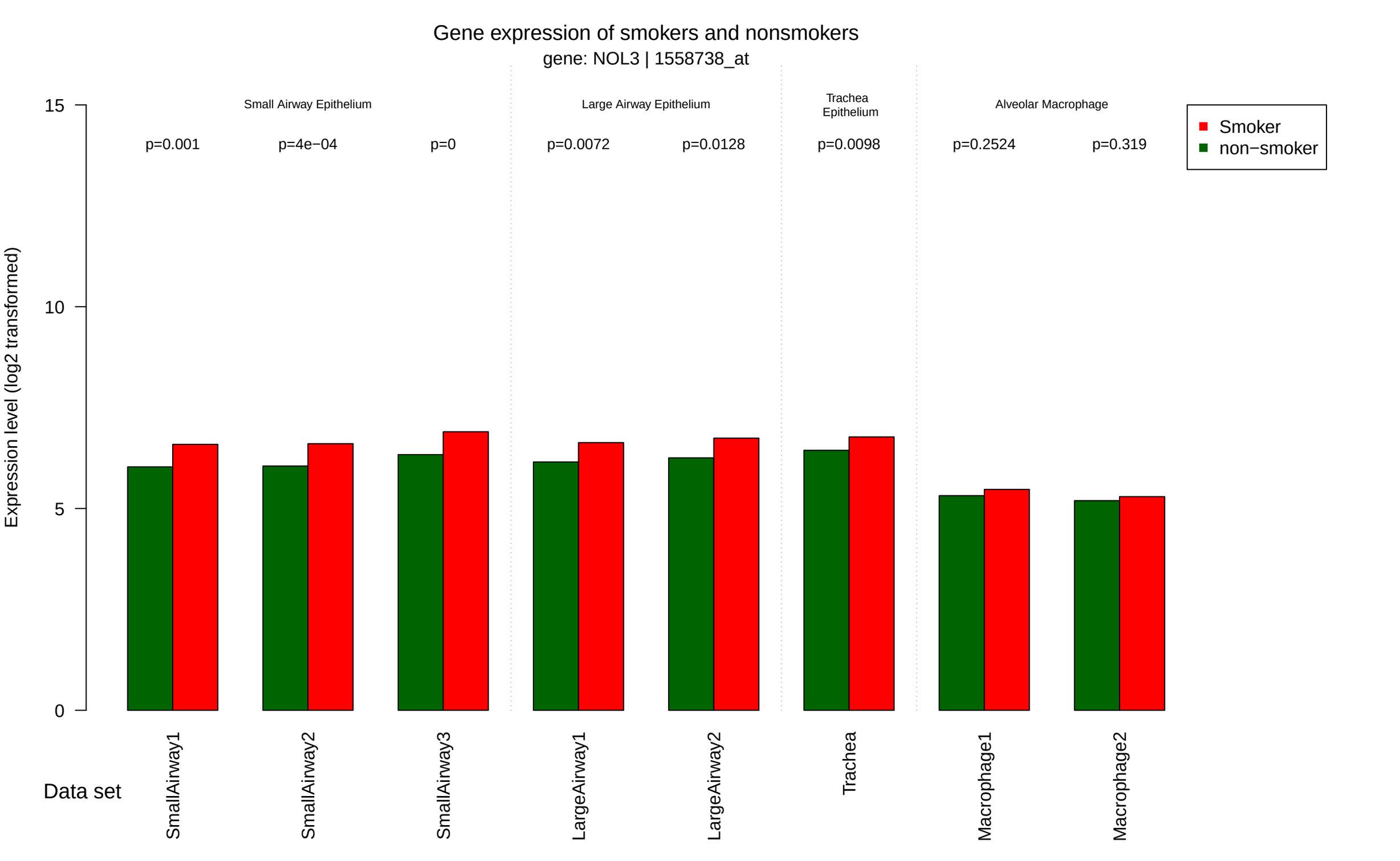
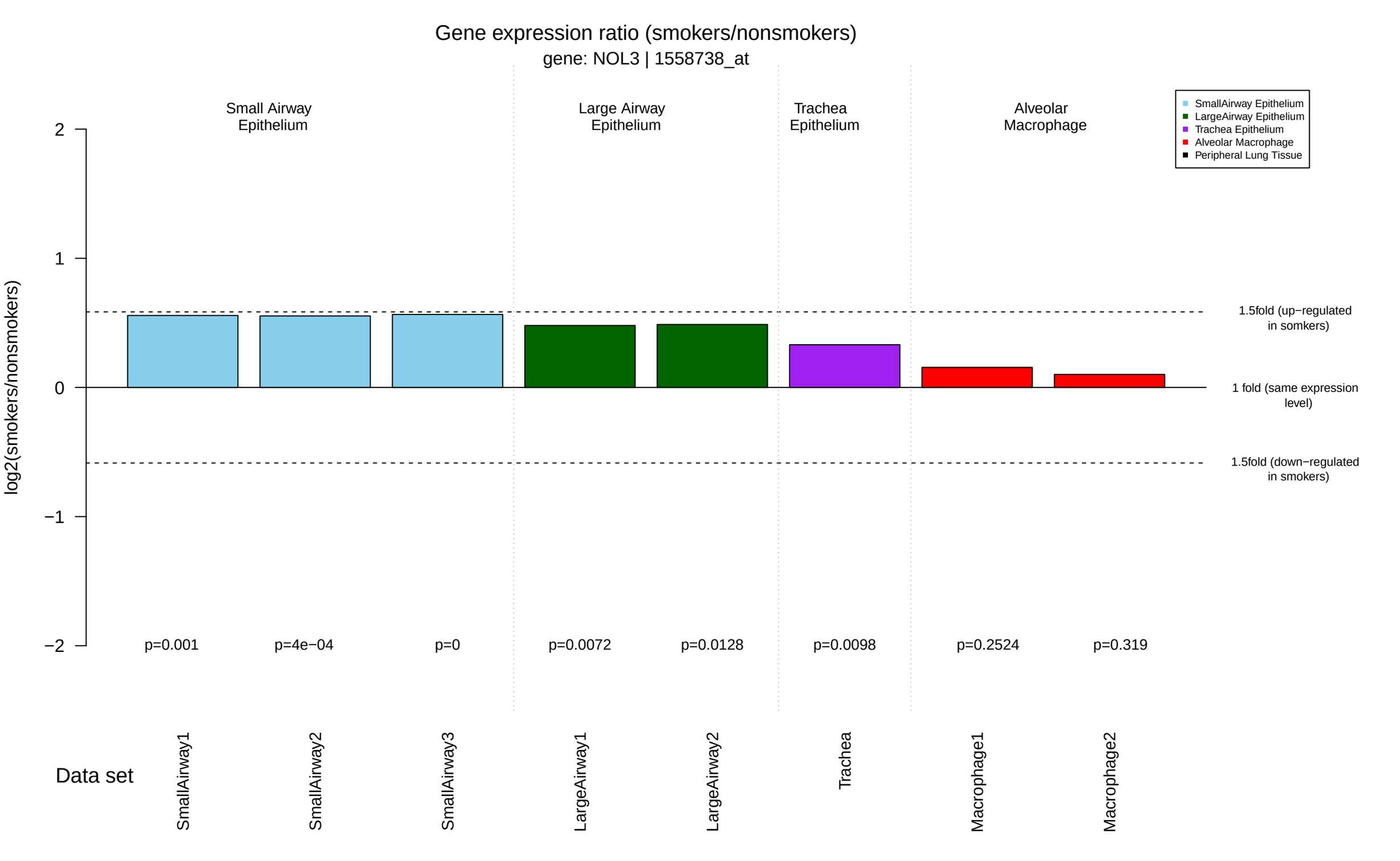
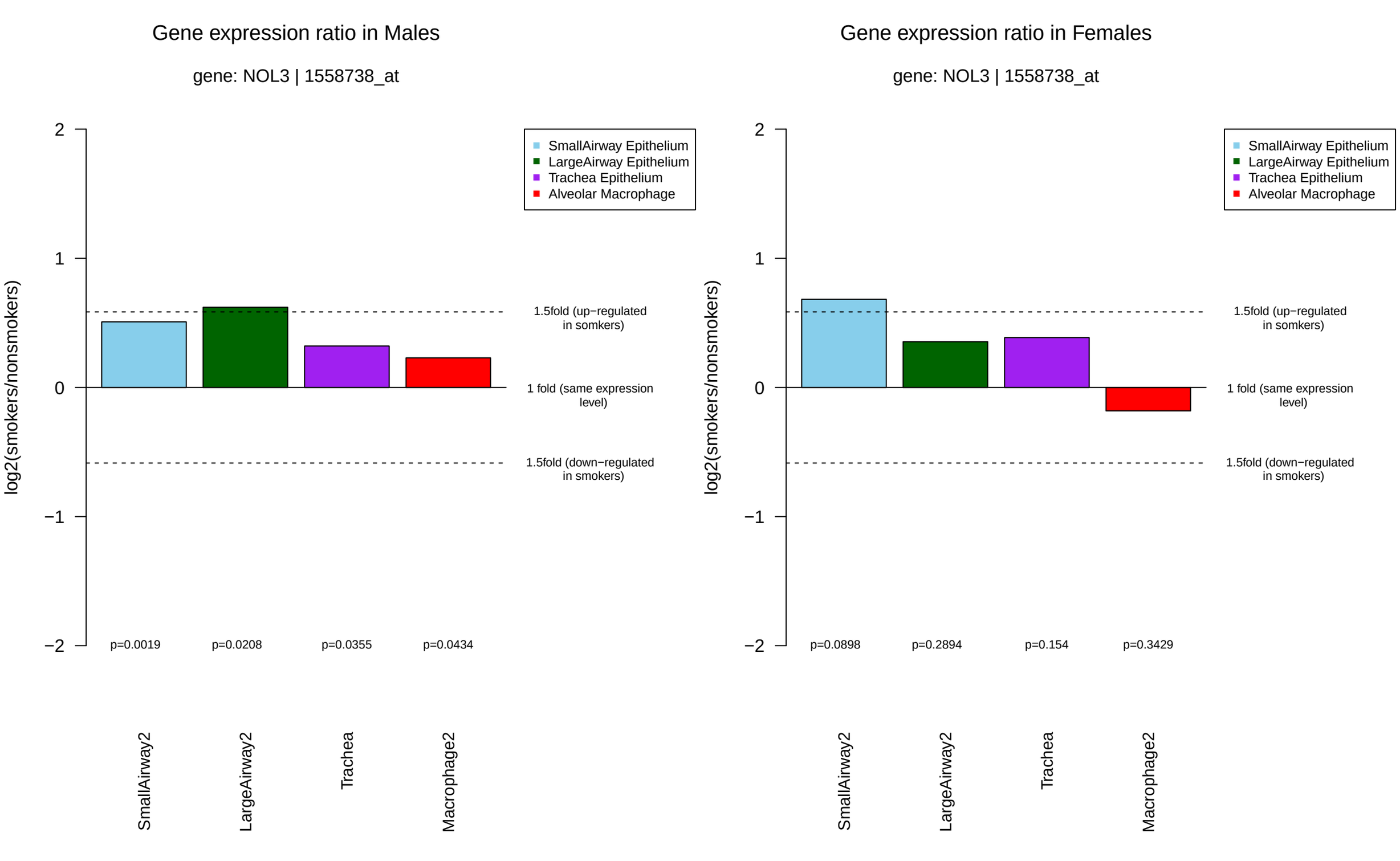
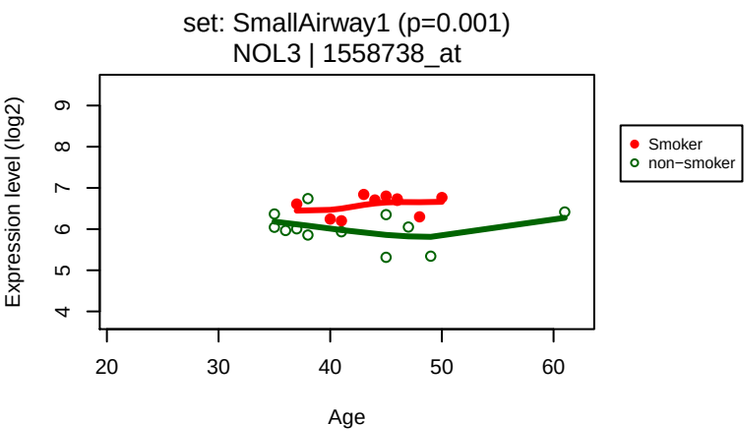
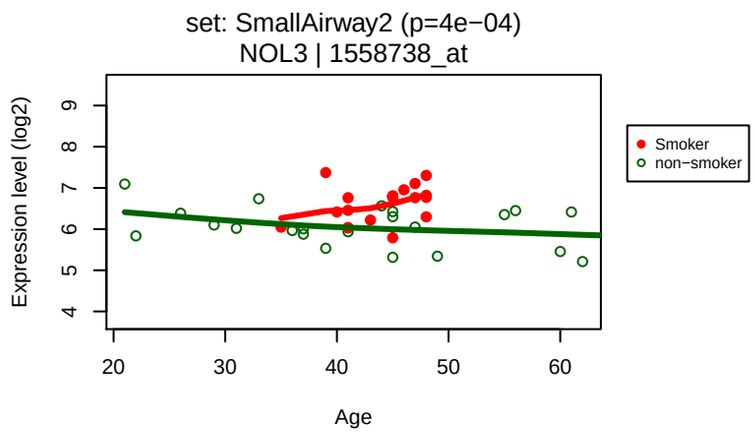
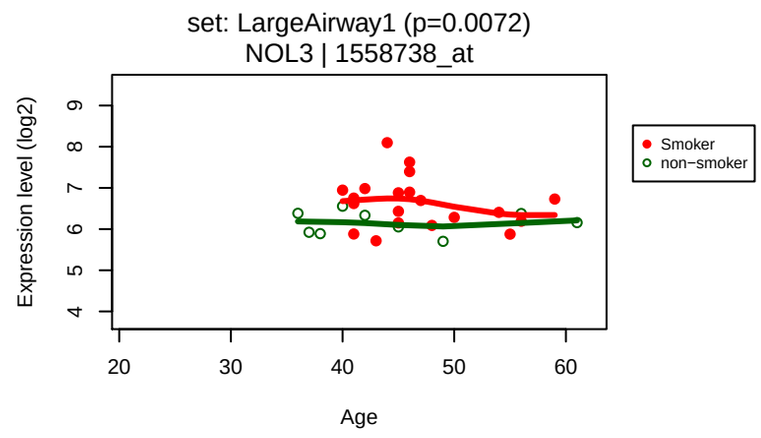

Gene expression of smokers and nonsmokers
gene: NOL3 | 1558738_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.001
p=4e−04
p=0
p=0.0072
p=0.0128
p=0.0098
p=0.2524
p=0.319
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

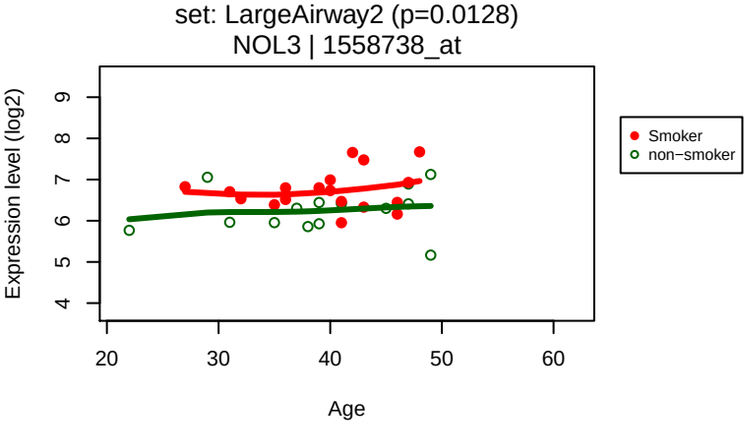

Gene expression ratio (smokers/nonsmokers)
gene: NOL3 | 1558738_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.001
p=4e−04
p=0
p=0.0072
p=0.0128
p=0.0098
p=0.2524
p=0.319
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

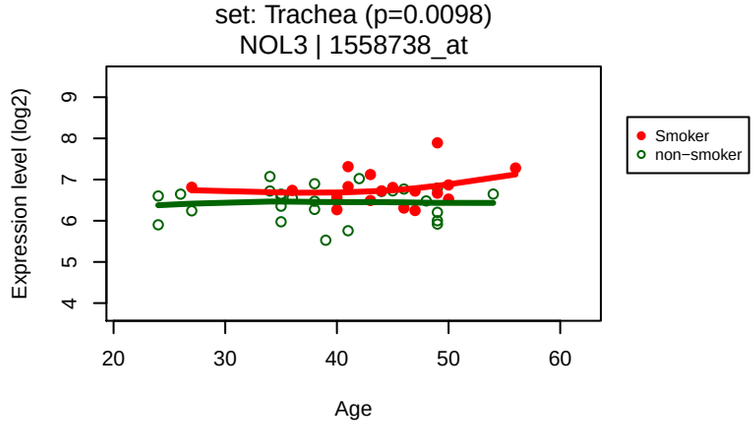

Gene expression ratio in Males
gene: NOL3 | 1558738_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0019
p=0.0208
p=0.0355
p=0.0434
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NOL3 | 1558738_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0898
p=0.2894
p=0.154
p=0.3429
SmallAirway2
LargeAirway2
Trachea
Macrophage2

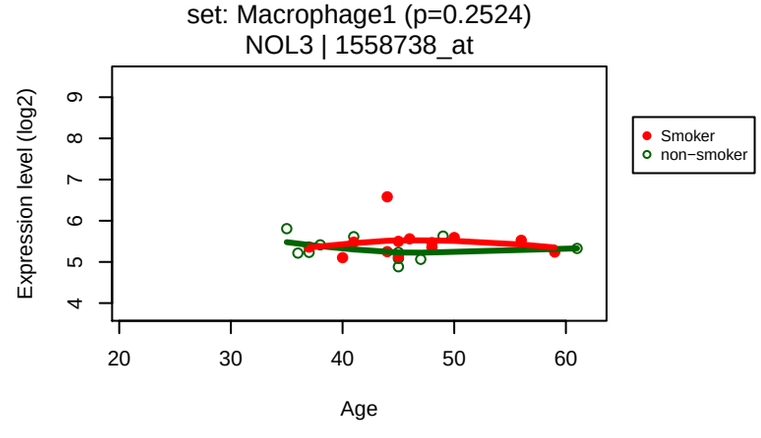

set: SmallAirway1 (p=0.001)
NOL3 | 1558738_at
Expression level (log2)
Age
Smoker
non−smoker

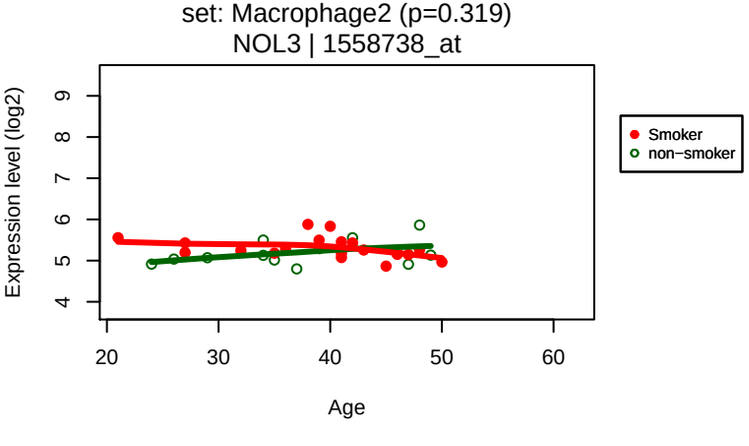

set: SmallAirway2 (p=4e−04)
NOL3 | 1558738_at
Expression level (log2)
Age
Smoker
non−smoker

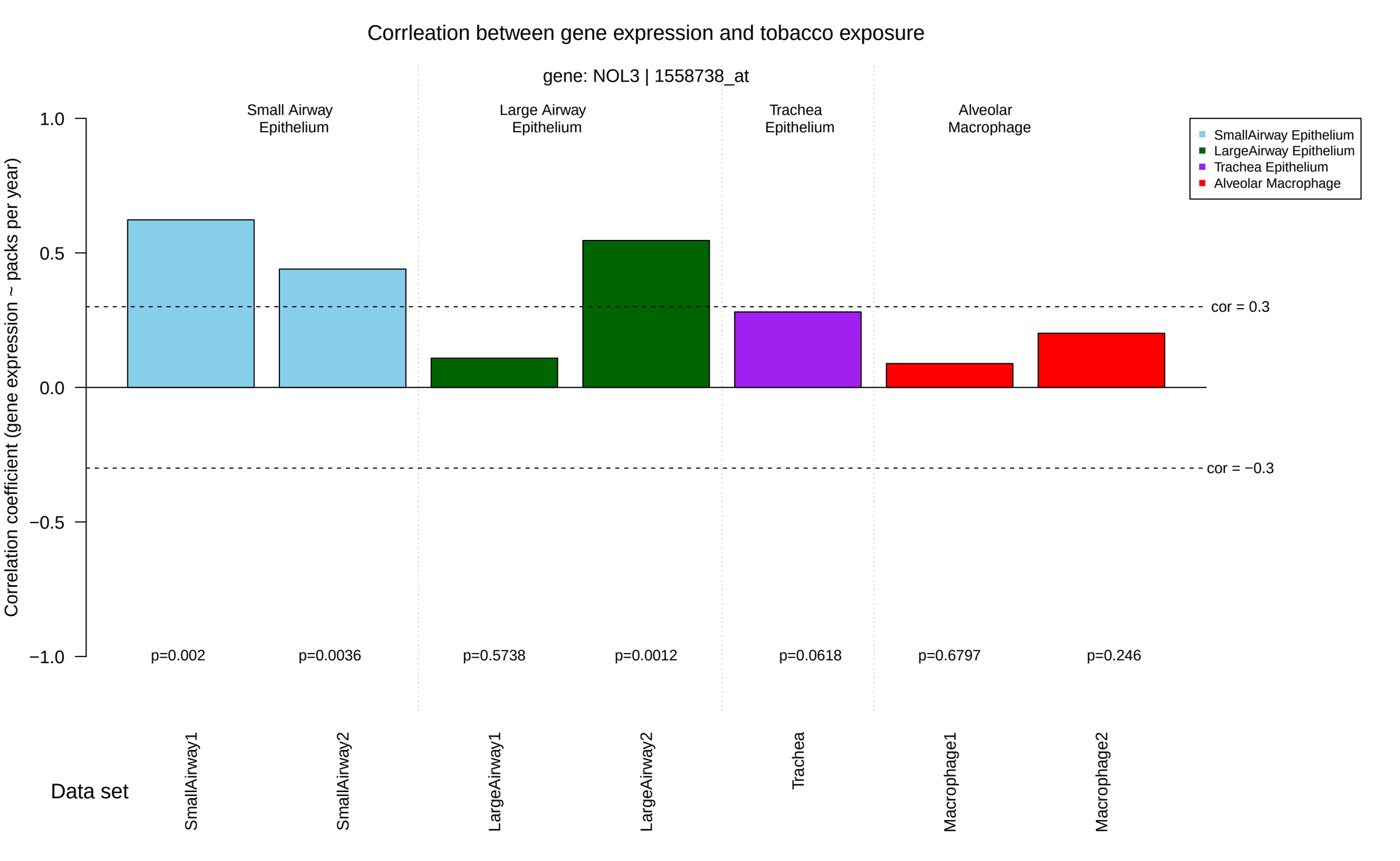

set: LargeAirway1 (p=0.0072)
NOL3 | 1558738_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0128)
NOL3 | 1558738_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0098)
NOL3 | 1558738_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2524)
NOL3 | 1558738_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.319)
NOL3 | 1558738_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NOL3 | 1558738_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.002
p=0.0036
p=0.5738
p=0.0012
p=0.0618
p=0.6797
p=0.246
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2